

Image-Based Video Game Asset Generation and Evaluation Using Deep Learning: A Systematic Review of Methods and Applications

Rafael Ribeiro , Alexandre Valle de Carvalho , and Nelson Bilber Rodrigues 

Abstract—Creating content for digital video game is an expensive segment of the development process, and many techniques have been explored to automate it. Much of the generated content is graphical, ranging from textures and sprites to typographical elements and user interfaces. Numerous techniques have been explored to automate the generation of these assets, with recent advancements incorporating artificial intelligence methodologies, such as deep learning generative models. This study comprehensively surveys the literature from 2016 onward, focusing on using machine learning to generate image-based assets for video game development, reviewing the deep learning approaches employed, and analyzing the specific challenges found. Specifically, the deep learning approaches employed, the problems addressed within the domain, and the metrics used for evaluating the results. The study demonstrates a knowledge gap in generative methods for some types of video game assets. In addition, applicability and effectiveness of the most used evaluation metrics in the literature are studied. As future research prospects, with the increase in popularity of generative AI, the adoption of such techniques will be seen in automation processes.

Index Terms—Diffusion models (DMs), evaluation metrics, generative adversarial network (GAN).

I. INTRODUCTION

THE rapid expansion of the digital video game (henceforward, video game) market, coupled with the relatively short-shelf life of individual games, has created an environment of relentless demand for innovation and new products.

Creating video game content is an intensive, expensive, and laborious task. It involves many experts, from visual artists to sound designers, all collaborating and iterating on a process to create and unify the many pieces that compose the envisioned product that is the final game. A considerable part of that process is graphical, ranging from textures and sprites to typographical elements and user interfaces.

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Rafael Ribeiro is with FEUP, 4200-465 Porto, Portugal.

Alexandre Valle de Carvalho and Nelson Bilber Rodrigues are with FEUP, 4200-465 Porto, Portugal, and also with INESC TEC, 4200-465 Porto, Portugal (e-mail: alexandre.valle@fe.up.pt).

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The idea to automate segments of this process is not new. As early as the 1980s, techniques such as procedural content generation, have been used to leverage the generation of game levels and assets. Despite this, the majority of visual elements in games are still crafted by hand. Machine learning techniques have garnered significant attention in the past years to accelerate the process of asset generation, namely, the use of generative adversarial networks (GANs) and variational autoencoders (VAE) [1] to generate game content with interesting results [2], as well as deep learning [3] and unsupervised learning techniques, such as Gaussian mixture models (GMM) [4]. This systematic review, implemented on the preferred reporting items for systematic reviews and meta-analyses (PRISMA) [5] framework, studies the problem of graphical asset generation for video games using machine learning methods, with emphasis in GANs, VAEs, and GMMs. Moreover, this study incorporates works based in diffusion models (DMs) and presents the promising results in image generation tasks. Furthermore, the contributions of this study are exposed in the following research questions.

RQ1: What trends can be observed in the popularity of deep-learning image-based generation techniques for video game content generation, and how have they evolved?

RQ2: What problems have been tackled in literature?

RQ3: What evaluation methods have been used?

The review supports that GANs have dominated the state-of-the-art image generation in the last five years and that DMs' popularity has spiked in the past year.

On the topic of graphical game content generation problems, it identifies 3-D assets, map or level, and image manipulation tasks as the most studied and game interface and logo generation as the least studied.

Concerning the evaluation methods, it classifies the most found metrics in the literature as state-of-the-art or not and as objective, perceptive, or hybrid, that is, if they measure known statistical properties, qualitative features, or a mixture of both. The leading indicators found for objective metrics are Fréchet inception distance (FID), peak-signal-to-noise-ratio (PSNR), and inception score (IS), and for perceptive metrics are structured similarity indexing method (SSIM), CLIP score (CS), CLIP-R-precision, and neural image assessment (NIMA). In addition, it expands upon qualitative means of evaluation, such as user studies and surveys.

TABLE I
SEARCH KEYWORDS

A	B	C
Image generation	Videogame	GAN
	Video game	GANs
	Video-game	GAN
	Videogames	GANs
	Video games	Diffusion
	Video-games	VAE
	Gaming	VAEs
		VAE
		VAEs
		Gaussian Mixture
		GMM
		GMMs

The study highlights two pitfalls in the game research community: the lack of shared datasets, which complicates reproducibility, and the insufficient ethical disclaimers in the description of experiments.

The rest of this article is organized as follows. Section II describes the steps to perform the systematic review based on PRISMA, and Section III details the execution of the chosen methodology. Section IV presents the review results, and Section V elaborates on the discussion about the achieved results. Finally, Section VI concludes this article.

II. REVIEW METHODOLOGY

The methodology and guidelines used to conduct the systematic review followed the PRISMA (PRISMA 2020) [5]. The following steps were conducted: search in scientific digital databases, define inclusion and exclusion criteria, gather data, filter, screen, and study the eligible papers in depth.

The search process to support the research questions was conducted through an exhaustive search and query process by consulting the following digital libraries of scientific literature, namely, ACM Digital Library, IEEE Xplore, Science Direct, Springer, CiNii, and arXiv. These were selected according to their relevance to the theme and the quality of publications.

A thesaurus of keywords is compiled and organized into Table I. A single keyword may contain a set of words, e.g., “VAE” is considered a single keyword, as opposed to “variational” and “autoencoder.” Each search string is composed of three operands, each coming from a distinct column, and joined using the logical AND operator. By their own turn, each operand is an expression containing all the keywords in a cell joined using the OR operator.

Columns A to C were chosen from broad topics such image generation narrowing down to specific techniques relevant to our study, namely, GANs, VAEs, and GMMs, having video games as the intermediate application area. Column B also narrows down to our focus application area, which is video games.

A. Inclusion and Exclusion Criteria

Books, conference papers, and journal articles are considered in the inclusion criteria for the digital library searches. Exclusion criteria are applied to the collection resulting from the previous search process to narrow down the works to screen: duplicate

TABLE II
QUALITY ASSESSMENT QUESTIONS

Index	Question	Answers
QA1	Did the work extend the state of the art on image generation techniques?	1. The work extends the state-of-the-art image generation techniques. 2. The work shows some new results, using existing techniques from the state of the art. 3. The work uses an existing technique and shows no new findings.
QA2	Did the work extend the state of the art on evaluating generated images?	1. The work introduces new evaluation metric(s). 2. The work applies commonly-used metrics, qualitative comparisons or user studies. 3. The work performs no explicit image evaluation.

titles, no available full-text access, citation types for “workshop,” “panel,” “tutorial,” “review,” “survey,” “proceedings,” or “editorial,” and nonEnglish titles or abstracts (checked using langdetect).¹ An exception was considered, namely, the seminal paper about GANs dating from 2014 [6]. Furthermore, we limited to papers starting in 2016.

B. Data Collection and Evaluation

The filtering and analysis of the collected works required extracting and processing the metadata (title, authors, publication year, publication type, and bibtex reference) and content-related information (application domain, problem, deep learning architecture, and image quality metric). The former is collected on retrieval, and the latter is extracted from the full texts in the last phase of the review.

C. Data Evaluation

For each included study, an assessment composed of the questions outlined in Table II is applied. For each question, the possible answers’ meaning is good, average, or poor, in this order.

III. REVIEW EXECUTION

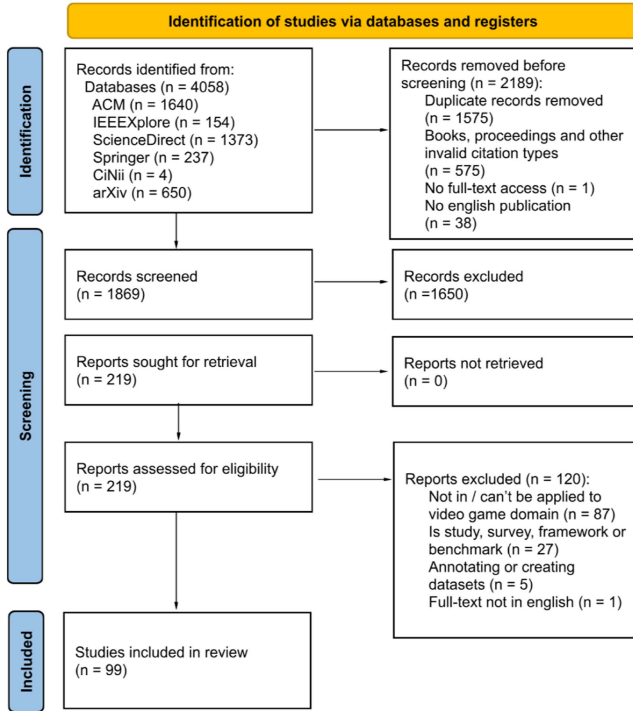
This section describes the steps taken to apply the previously described methodology.

A. Querying and Filtering

Query strings were generated from keywords and aggregation rules (see Table I) and executed in digital libraries. Response pages were scraped for titles, abstracts, publication years, and types, then saved in a datasheet. The interaction pipeline with the databases’ web interfaces was implemented using Selenium WebDriver.² The datasheet was filtered based on the exclusion criteria (see Section II-A).

¹[Online]. Available: <https://pypi.org/project/langdetect/>

²[Online]. Available: <https://www.selenium.dev/>

Fig. 1. PRISMA flow diagram. ⁴

B. Screening and Eligibility Assessment

Titles and abstracts from the previous stage were analyzed to trim down the included works. These were liberally included when it was believed they could be reasonably applied to the domain of video games, meaning, where the research or research-resulting prototypes could be transferred or applied to a channel of video-game development. Works where this was not found to be the case, and were thus excluded, featured: application domains in the medical, fashion, industrial, traffic management, or city planning; denoising/dehazing, data prediction, image retrieval, steganography, optical character recognition; facial expression, emotion, speech, and music generation or identification.

The full texts of the remaining works were assessed, resulting in 99 included articles. Abstracts, contexts, conclusions, and experimental results were analyzed to determine relevance. Non-relevant works were excluded with reasons annotated. Citations for each work were collected in BibTeX format using a Google Scholar API.³

IV. REVIEW RESULTS

Fig. 1 shows the resulting PRISMA flowchart, with the number of papers retrieved and excluded at each stage and for what reason.

A total of 4058 papers were first identified after querying all databases; 1869 remained after the automatic application

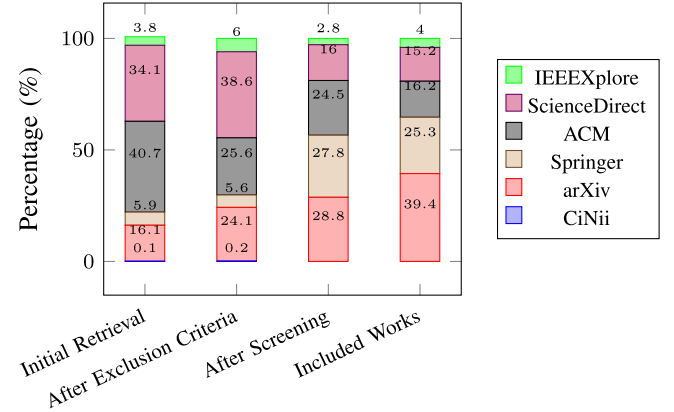


Fig. 2. Distribution of papers retrieved from digital libraries at each step of the systematic review process.

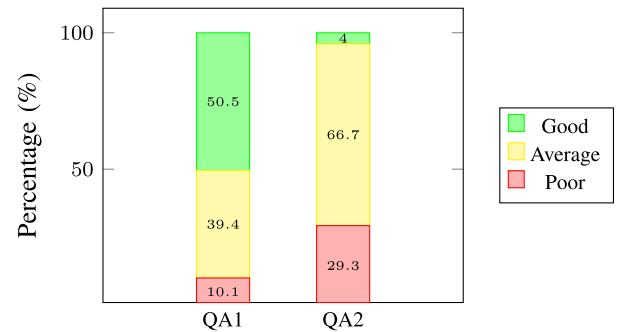


Fig. 3. Quality assessment answers distribution.

of the inclusion and exclusion criteria, 1650 were excluded after the screening, and 120 were excluded after the eligibility assessment. The reasons for this last round of exclusion, besides the ones listed in Section III-B, are mainly false positives from the screening and prescreening stage, as most are cases outside the video game domain or of undesired publication types, such as surveys and studies. The final number of included works is 99.

Fig. 2 illustrates the distribution of papers from each digital library at each review step. Despite representing just 16.1% (650 of 4058) of the initial retrieval, in the end, the major contributor to the final pool is arXiv, with 39.4% (39 of 99). The second major contributor is Springer, with 25.3% (25 of 99), followed by ACM, with 16.2% (16 of 99), ScienceDirect, with 15.2% (15 of 99), and IEEE, with 4.0% (4 of 99). CiNii is the only digital library that did not contribute to the final pool of included works, with 0.1% and 0.2% in the first two stages, respectively, and no remaining papers after screening. On an important note, 13 of the 39 arXiv papers, at the time of retrieval, were in preprint status and not yet peer-reviewed.

On the quality of the findings, Fig. 3 shows the distribution of the answers to the quality assessment questionnaire.

Only 10.1% of the works present no new image generation results or techniques. 50.5% present novel techniques, and 39.4% present new results using existing techniques. The latter mostly

³[Online]. Available: <https://github.com/ckreibich/scholar.py>

⁴[Online]. Available: <http://prisma-statement.org/prismastatement/flowdiagram.aspx>

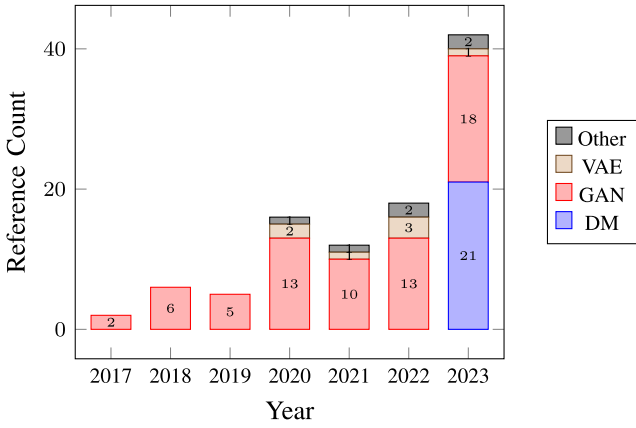


Fig. 4. Included papers and applied image generation techniques' distribution per year (2017–2023).

represent combinations of two or more existing approaches to a new domain or problem [7], [8].

In total, 29.3% performed no image evaluation of the results. In the majority of the cases, this is because the work itself does not target image quality but other aspects of video game quality, such as level novelty, although a small percentage of this fraction actually performed no evaluation at all. The majority of works (66.7%) performed adequate evaluations, be it through quantitative measurements, qualitative comparisons and user studies, or a combination of both. A small fraction (4%) developed new metrics for the specific problem at hand [e.g., Ho et al. [9] proposed “F-score of Inception and SImilarity (FISI), which compromises FID and SSIM to focus on both target features and retaining input content”].

V. DISCUSSION

Further analysis is devoted to the findings of the previous section by answering the research questions proposed in Section I.

A. Answering Research Questions

RQ1: Of the 4058 originally retrieved documents containing mentions of image generation, only 99 were included. These implemented a myriad of different approaches, from level-generating cellular automata [10] and reinforcement learning agents [11], [12] to feedforward neural networks [13] and generative playing networks [14].

Fig. 4 shows the chronological distribution of deep learning solutions in the included works. GANs are the most prevalent, representing 67.7% of the total works' approaches. However, in this past year, DMs have been gaining popularity. Representing just 21.2% of the full works share but also 50.0% of this past year's works, these networks have shown great promise in image generation [15]. One interesting observation is that, despite being included in the original search terms, no GMM-based works passed the screening process, nor did works published in 2016.

RQ2: Video game development pipelines contain many problems that can be solved using image generation. Included works

TABLE III
IMAGE MANIPULATION SUBPROBLEMS AND INPUTS

Task	Inputs	References
Colorization	Sketch outline	[79]
Semantic image editing	Input image + text instruction	[71], [72]
Stylized image generation	Text prompt	[69]
	Style image + structure image	[73]
	Text + Sparse semantic map	[80]
	Content image + prompt + style image	[74]
Content image generation	Background image + foreground content text	[76]
Photograph stylization	Photograph	[9], [75], [81]
Improvements	Input text generation query	[7]

were first categorized by their application domain, as can be seen in Fig. 5. Then, the focus is directed to the research scope, and a selected domain is analyzed to identify specific tasks.

An emphasis on level/map generation is evident. The segment of these works focused on solving the task of 2-D level generation in platformer [14], puzzle [89], and role-playing [13], [70], [90] games, from both game design and visual aesthetic perspectives. In contrast, another focused on the 3-D generation of world terrains [16], [17], [94], [95], indoor scenes [8], [21], and urban settings [39].

Besides maps, 3-D content generation is a main trend and it has been applied to model texturing [32], hair [35], asset [33], [34], [37], voxel shape [31], and human face/head [18], [19], [26], [27], [38], [41] generation. Another main area of focus is Font/Typography Generation. Namely, the generation of artistic fonts from natural language descriptions [56], high-level attributes [66], handwriting [58], [61], [62], [65], and style transfer [63], [68]. Lastly, character concept generation from random noise [45], [51], [54], line drawings/sketches [46], [48], and blurred silhouettes [49]. Less explored generation problems include logos [103], pixel art characters [47], [48], [77], [90], sprite assets [13], [48], [90], map chunks [13], [90], input image pixelization [78], game effects [73], and graphical user interfaces (GUI) [86], [87], [88].

In summary, half of the works reviewed are applied to the following domains: 3-D content, level/map, image manipulation and font/typography.

RQ3: Assessing the quality of generated images is a crucial aspect of image generation model development. Measuring the degree of certain characteristics of results is essential to determine performance and achieve progress. Various methods have been developed to objectively and subjectively measure these synthesized images' fidelity, realism, perceptual accuracy, and creativity. The literature review found a total of 35 different objective computable metrics, along with quantitative evaluation methods, and to answer this research question, these were categorized and separated by the research problem formulation (the metrics in the first column of Table IV refer the citing paper; the absence of citation means no first literature reference found).

Evaluation metric can be classified according to one or more criteria as follows:

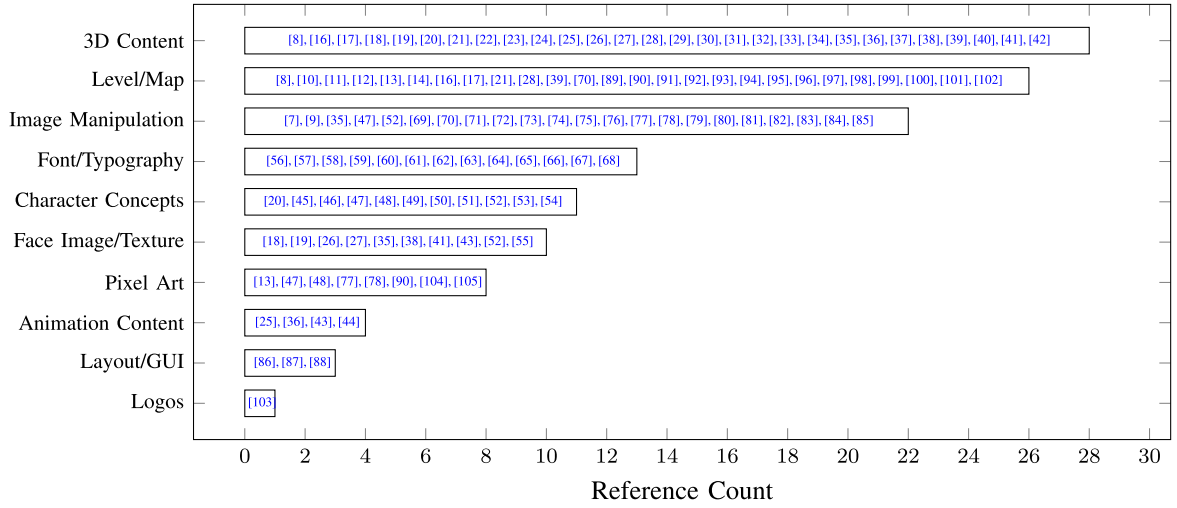


Fig. 5. Distribution of included works by application domain.

TABLE IV
IMAGE EVALUATION METRICS CATEGORIZATION

Evaluation Metric	References	Opinion-Aware	Alignment	Diversity	Fidelity	Full-Reference	No-Reference
FID [106]	[9], [17], [18], [20], [21], [28], [32], [33], [34], [35], [47], [57], [60], [61], [65], [66], [68], [69], [71], [75], [78], [80], [82], [83], [86], [87], [93], [95], [103]	x		x	ImageNet [109]		
FID-Ori and FID-3D [34]	[34]	x		x	Custom test shapes set		
IS [107]	[9], [28], [37], [53], [66], [68], [71], [76], [95], [103]	x		x	ImageNet [109]		x
Masked IS (m-IS)	[53]			x	ImageNet [109]		x
Conditional IS and CFID	[57]			x	Custom test shapes set		x
Gan-Train & Gan-Test [119]	[61]			x	Custom real image set		
MMD [120]	[31]			x			
Kernel Inception Distance (KID) [110]	[27], [32], [78]	x		x	ImageNet [109]		
Zero normalized cross-correlation [121]	[39]			x	Custom single image	x	
FISI [9]	[9]	x		x	ImageNet [109]	x	
Single Image FID [122]	[22]	x			Custom single image	x	
PSNR	[29], [38], [44], [46], [59], [79], [92], [95], [100]				Custom single image	x	
Mean squared error	[38], [39], [47], [59], [92]				Custom single image	x	
Root mean squared error	[38], [43], [48], [63]				Custom single image	x	
Mean absolute error	[43]				Custom single image	x	
Image Euclidean distance [123]	[52]					x	
SSIM [111]	[9], [29], [43], [44], [46], [47], [53], [59], [66], [68], [79], [92], [95], [100]					x	
MS-SSIM [115]	[29], [31]					x	
Masked SSIM	[53]					x	
Directional CLIP similarity [124]	[72]	x				x	
Visual information fidelity [125]	[46], [59]	x				x	
Learned perceptual image patch similarity [126]	[29], [66]	x			Custom single video/video still	x	
LAION-Aesthetic score (LAION-Aes) [127]	[74]	x	x		Custom dataset (BAPPS)	x	
(CS) [112]	[7], [13], [18], [27], [28], [69]	x	x		LAION-5B [127]	x	
CLIP-R-Precision [113]	[26], [30], [93]	x	x		WebImageText [128]		x
Semantic Object Accuracy [129]	[93]	x	x		WebImageText [128]		x
HPS [131]	[74]	x	x		COCO [130]		x
NIMA [114]	[7], [71], [76]	x			Custom Dataset		x
Blind/referenceless image spatial quality evaluator [132]	[73]	x			"AVA [116], TID2013 [117], and LIVE [118]"		x
Color histogram [133]	[79]						x
Entropy of images [134]	[46]						x
Aesthetic Evaluation [135]	[50]						x
Natural image quality evaluator [136]	[73]						x
Perception-based image quality evaluator [137]	[73]						x
Cumulative probability of blur detection [138]	[43]						x

- 1) metrics that leverage human scores and opinions in some way within a process (opinion-aware);
- 2) metrics that measure alignment or accuracy between an input and some description of its contents (be it through text prompts, layout maps, or others);
- 3) metrics that evaluate diversity of the model's outputs;
- 4) metrics that reward input fidelity or likeness towards some other set of images (directly, or indirectly through a trained predictor);

- 5) metrics that measure an input against a pristine reference (full-reference);
- 6) metrics that leverage only an input's inherent statistical features (no-reference).

The most common metrics referenced in the literature are FID [106], PSNR, and IS [107]. PSNR evaluates the quality of an image or video by measuring the difference between the original and compressed versions. IS is maximized when the entropy of the distribution of labels predicted by the Inceptionv3 [108]

model for the generated images is minimized and when the predictions of the classification model are evenly distributed across all possible labels, indicating both “sharp” or “distinct” images and diverse outputs from the generative model. However, it evaluates only the distribution of generated images, unlike FID, which uses the same predictor but also considers real images from the training set. It is calculated by measuring the distance between feature vectors calculated for real and generated images. The Inceptionv3 [108] predictor, along with its predecessors, was trained on the ImageNet [109] dataset, which carries a known set of biases that are further described in Section V-B.

Variations of these metrics have been implemented to better evaluate a solution on a case-by-case basis. For example, masked IS (m-IS), a variation of IS, masks out the image backgrounds before evaluation. Kernel inception distance (KID) [110], which uses the maximum mean discrepancy (MMD) to measure distribution dissimilarity, and conditional FID (CFID), which measures the spread of conditioned classes over actual classes, are variations of FID.

The most referenced perceptive metrics in the performed literature review are SSIM [111], CS [112], CLIP-R-Precision [113], and NIMA [114]. SSIM is a traditional measure of the similarity between two images. It is based on the concept of structural information, incorporating critical perceptual phenomena, such as luminance masking and contrast masking terms. Its modern variation, multiscale structural similarity indexing measure (MS-SSIM) [115], performs the evaluation at multiple scales through a multistep downsampling process. CS and CLIP-R-Precision are used to evaluate the correlation between a generated caption for an image and the actual content of the image and are highly correlated with human judgment. A variation of these, directional CLIP similarity, measures the consistency of the change between two images and two image captions. NIMA uses a 1-D “aesthetic quality” metric based on training a machine learning model to match human judgements of visual value. The model was trained on subsets of various datasets (AVA [116], TID2013 [117], and LIVE [118]).

Despite the variety of existing metrics, many visual qualities are hard to quantify objectively and computationally, that is, calculate numerically through algorithmic means. This fact is especially accentuated in the context of entertainment experiences such as video games, where the visual medium serves multiple purposes besides static displays of images, and so measuring their effectiveness becomes too complex a task. Many authors advocate the use of user studies [139], surveys, focus groups, or expert panels as alternative qualitative evaluations, that is, resorting to human judgement to assess the results’ quality. Table V list the studies and works on the survey that use qualitative methods for evaluation.

B. Considerations About Datasets

Issues about the data heterogeneity availability of datasets for training have been tackled in the literature by [141]. It was observed that the problem remains. There is no standardization in the datasets used to perform the experiments. Most of the

TABLE V
WORKS WITH NO OR QUALITATIVE EVALUATIONS

Evaluation Metric	References
Qualitative Comparison	[9], [18], [19], [23], [24], [27], [30], [41], [46], [50], [53], [54], [61], [63], [65], [67], [69], [76], [78], [80], [81], [88], [92], [94]
User Study	[9], [25], [27], [46], [49], [55], [65], [70], [77], [81], [87], [88], [95], [97]
None	[8], [10], [11], [12], [14], [16], [40], [45], [51], [56], [58], [62], [64], [89], [90], [91], [96], [99], [140]

experiments rely on custom datasets, and the ordinary practice of data sharing with the research community still needs to be improved. Nevertheless, the following games and datasets were used more than once: *Pokémon* [45], [47], [54], *Sokoban* [12], [91], and *COCO* [23], [23], [69], [71], [80], [93].

With the advent of generative techniques, it was necessary to prepare and use datasets with a large amount of data to feed the complex artificial algorithms. This raised issues related to privacy, copywriting and authorship of the data [142]. The explosion of generative AI ethics and privacy concerns were inflamed, and the current laws from the European Union, and the US have trouble keeping up with the novelty of the field [143]. The digital gaming community has highlighted issues related to obtaining data through large-scale, indiscriminate web crawls conducted without the consent of data owners [144]. This situation raises significant concerns regarding authorship and plagiarism, particularly in relation to the provenance of data used to construct datasets and the copyright implications of content generated based on these foundational datasets [145]. In this review, only four explicitly refer to ethical issues [23], [34], [55], [56].

VI. CONCLUSION

The growing trend of using artificial intelligence to generate content for games inspired the survey, with a focus on studying the techniques used by the scientific community from 2016 to 2023.

PRISMA was the methodology chosen to conduct the systematic review. After filtering and screening, 99 papers were selected to answer the research questions related to the approaches to generating synthetic images, the associated problems and challenges, and how the evaluation metrics have been applied.

The results extracted by the literature review demonstrate that GANs and DMs are the most referenced techniques, and their popularity has increased over the years. Also, it was possible to verify that these techniques are popular in scientific literature to address problems, such as 3-D content, level/map generation, and image manipulation. We list the qualitative and quantitative metrics used for evaluation of generated content. Qualitative analysis was mainly performed using user studies, survey studies and interviews. Nevertheless, the focus of this study was quantitative metrics, which were characterized in terms of six fields: opinion-awareness. FID, PSNR, and IS are the leading indicators for objective metrics. For perceptive metrics, SSIM, CS, CLIP-R-Precision, and NIMA are the most referenced techniques.

Some prospects for future research lie in segments of the game content development pipeline. Namely, layout generation and logo generation were the areas with the least included works identified. Other areas not referenced in this study and whose generation were considered relevant include nonhuman 3-D models (vehicles, animals, tools, etc.), special effects, particles, postprocessors, shaders, and other unconventionally-rendered game content.

It is important to note that ethical and copyright concerns are mainly absent in most articles reviewed, highlighting the need for greater awareness among researchers and practitioners.

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